Inter-University Avishkar Research Convention: 2026-27

Slot No.: _____

Category: Engineering & Technology

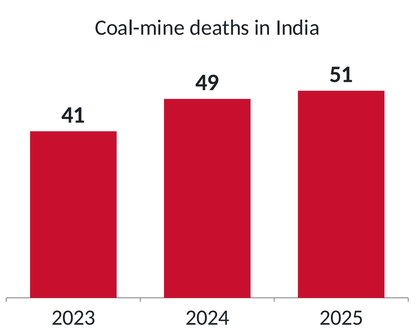
Level: UG

MineSafe

Intelligent Safety Gear Compliance, Health Monitoring and Navigation System for Underground Miners

- No GPS
- No internet
- No new cabling

01 THE PROBLEM



141 deaths, 2023–25

355 serious accidents

0 GPS or phone signal below ground

In every rescue: nobody knows who is underground, where, or whether they are alive.

02 OBJECTIVES

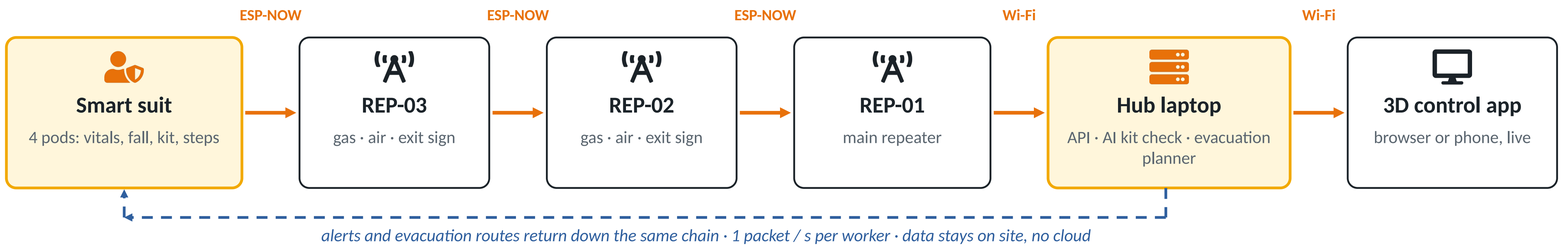
- **Kit compliance** — RFID / ID card + YOLOv8 photo check at the gate
- **Health** — heart rate, SpO_2, temperature, falls, no-movement
- **GPS-free location** — steps + heading, corrected at every repeater
- **Self-healing network** — hop-by-hop ESP-NOW, no cabling or internet
- **Gas and heat** — CH_4, CO, H_2S, wet-bulb at every repeater
- **Low cost** — < ₹7,000 / repeater · ≈ ₹12,800 / suit

03 RESEARCH GAP

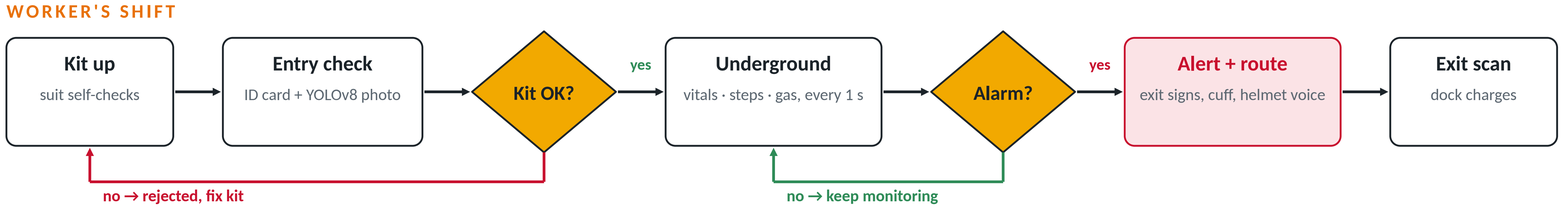
System	Kit	Vitals	Fall	Location	Gas	Price
MineGuard 360	✗	✓	~	~	✓	quote
MineARC GuardIAN	✗	✗	~	zone	✓	quote
Leantegra UWB	✗	✗	✗	✓	✓	quote
Rajant Health	tag	✓	✓	~	~	quote
SmartKavach	✗	~	✓	✗	✗	quote
36Zero + Bodytrak	~	✓	✓	✗	✗	quote
MineSafe	✓	✓	✓	✓	✓	**₹12.8k**

Novelty: the first low-cost system that combines kit check, vitals and GPS-free location in one self-healing network.

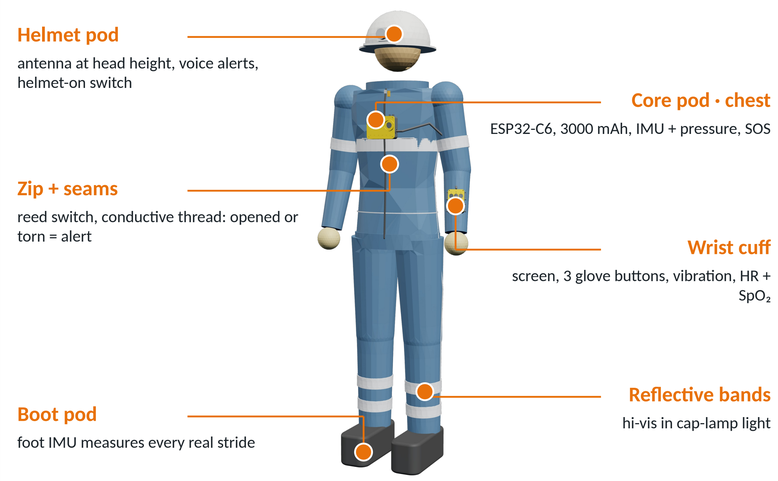
04 METHODOLOGY — ARCHITECTURE AND DATA FLOW



WORKER'S SHIFT



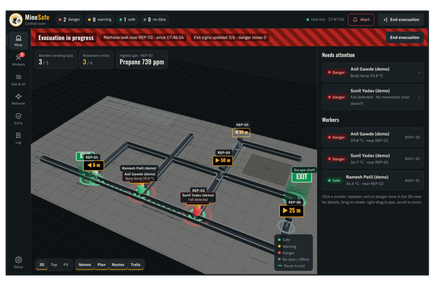
05 V4 SMART SUIT



Never locked: removal is detected, never prevented

06 LOCATION WITHOUT GPS

1. **Count steps** — stride × gyro heading; pressure gives the level
2. **Snap to tunnel** — position pulled onto the mine-map centre line
3. **Reset at repeater** — signal ≥ −55 dBm → drift back to zero
4. **Route around danger** — Dijkstra from all exits, hazard zone ×1000 cost



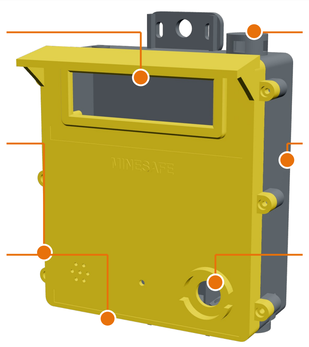
Live 3D control app: routes bend around a gas leak, every exit sign updates

07 REPEATER: BUILT FOR THE MINE

LED exit sign — arrow + metres, under a drip hood

85 dB buzzer — works if the hub is down

CO / H_2S below — heavier gases; flame arrestors

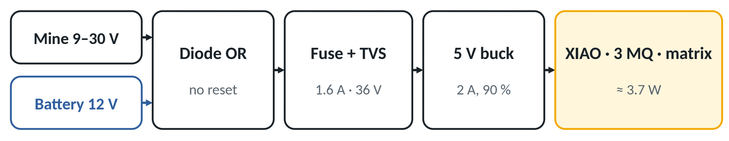


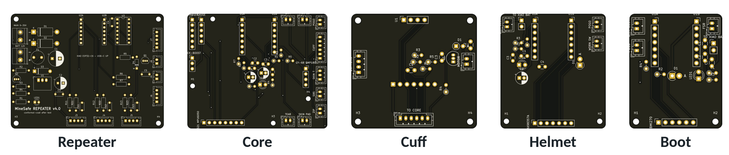
CH_4 port on top — methane rises; rain chimney

IP65 gasket — silicone cord, 6 × M4

Guarded SOS — glove-sized, knock-proof

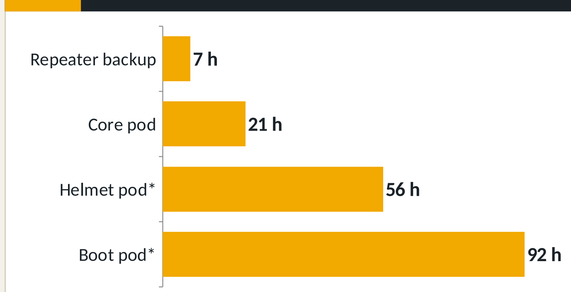
08 ELECTRICAL DESIGN AND PCBS





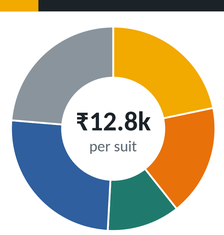
5 own 2-layer boards · KiCad + Freerouting · 100 % routed · 0 unconnected pads

09 ENERGY BUDGET



*Hours on one charge; a shift is 12 h. *duty-cycled radio. Repeater ≈ 3.7 W on mine power.*

10 COST



Boot pod	3,270
Fabric + sensing	3,040
Core pod	2,790
Helmet pod	2,260
Wrist cuff	1,480

₹6,862 / repeater · ₹3.26 L for 10 repeaters + 20 suits

11 VALIDATION

Test	Pass criterion	Status
PCB DRC, 5 boards	0 unconnected pads	Passed
CAD fit, 5 cases	no collisions	Passed
Power budget	core pod ≥ 12 h (21 h)	Passed
Repeater chain	self-forms, 3 / 3	Passed
Gas + local buzzer	alarm at limit	Passed
Distance error	≤ 5 % over 100 m	In progress
Alert latency	≤ 3 s for 95 %	Planned
Fall detection	≥ 90 %, ≤ 1 false / h	Planned

12 CONCLUSION

Every worker **checked, monitored, located and reachable** for the whole shift, on hardware a small mine can install itself.

- Working V3 prototype: 3 repeaters, body unit, hub, 3D app
- V4: 5 PCBs, 5 mine-rated cases, smart suit
- Next: LoRa backbone, certified gas sensors, mine trial

SDG 3 Good health · SDG 8 Decent work · SDG 9 Industry

References [1] Ministry of Mines data, ETV Bharat, Aug 2026 · [2] Coal Mines Regulations 2017, reg. 153 · [3] NIOSH Pocket Guide to Chemical Hazards · [4] Bandyopadhyay et al., Wireless Information and Safety System for Mines, JSIR 2009 · [5] IMU-assisted UWB positioning in underground coal mines, Micromachines 2023 · [6] Real-time PPE compliance detection (YOLO), Sustainability 2023